

- 1. List 3 common hacking methodologies (such as phishing, brute force, SQL injection) and for each, write one real-world example of how it could be used to compromise an account on an app like Instagram or Paytm.**

Phishing

- An attacker creates a fake login page or sends a deceptive message to trick a user into revealing their username, password or OTP.
- Instagram – An attacker sends a fake “Your account will be suspended” message containing a fake Instagram login link. If the victim enters their credentials the attacker can use them to attempt access to the account.

Brute Force

- An attacker repeatedly tries many possible passwords until the correct password is found.
- Paytm – If a user has a weak commonly used password and the service lacks effective login protections such as rate limiting or account lockout an attacker could repeatedly guess passwords to try to gain access.

SQL Injection

- An attacker exploits an improperly secured application by inserting malicious SQL input into a field potentially allowing unauthorized access to database information.
- Instagram – If an application’s login system directly places user input into a database query without proper validation and parameterization. A valuable login from could potentially be manipulated to bypass authentication or expose account-related data.

2. Simulate a simple brute force attack by writing a Python script that tries all 4-digit PIN combinations (0000-9999) to guess a preset password variable. Print the number of attempts taken to find the correct PIN.

- **Here is Python script to guess 4-digit PIN combinations:**

```
#Simple Brute Force Simulation

#Tries every 4-digit PIN from 0000 to 9999

Preset password = "4827"

Attempts = 0

For number in range (10000):

    Pin = f"{number:04d}"

    Attempts += 1

    If pin == preset_password:

        Print("PIN found!")

        Print("Correct PIN:", pin)

        Print("Number of attempts:", attempts)

        Break
```

Output:

If the preset password is 4827:

PIN found!

Correct PIN: 4827

Number of attempts: 4828

- 3. Research and describe the steps involved in a phishing attack targeting users of a popular food delivery app like Zomato or Swiggy. Include at least 4 steps and explain what the attacker and victim do at each stage.**

Phishing attack targeting a Food-delivery App User

- A phishing attack is a type of social-engineering attack in which an attacker impersonates a trusted company to trick a victim into revealing sensitive information or performing an unsafe action. Food-delivery platforms such as Zomato and Swiggy can be impersonated through fake support messages, refund offers or suspicious links. Zomato specifically warns users not to share OTPs, CVVs UPI PINs or card details with people claiming to represent Zomato.

1) Attacker prepares the fake message

- **Attacker:**
- The attacker creates a convincing message pretending to be from Zomato or Swiggy. For example, they may claim that an order has been cancelled, a refund is pending or the user's account needs verification. The message may contain a malicious link or direct the victim to a fake customer-support channel.
- **Victim:**
- The victim receives the SMS, WhatsApp message, email or social-media message and may believe it is a genuine communication from the food-delivery company.

2) Attacker creates a sense of urgency

- **Attacker:**
- The attacker tries to make the victim act quickly by saying things such as "Your refund will expire" or "Your account will be blocked unless you verify it". This reduces the chance that the victim will carefully check the message.
- **Victim:**
- The victim becomes concerned about losing money, an order or access to their account and decides to follow the instructions.

3) Victim visits the fake website or contacts the fake support channel

- **Attacker:**
- The attacker directs the victims to a fake webpage or fraudulent “customer support” contact that looks legitimate. The page may imitate the appearance of the real service.
- **Victim:**
- The victim may enter information such as their phone number, login information or other requested details. In a fake-support scam, the victim might also be persuaded to provide a verification code.

4) Attacker attempts to obtain sensitive information or payment

- **Attacker:**
- After gaining the victim’s trust. The attacker may request an OTP, CVV, UPI PIN, card information or another sensitive credential. The attacker may falsely describe this as necessary for a refund or account verification.
- **Victim:**
- If the victim provides the information the attacker may use it for account takeover or financial fraud.

4. Given the following fake login page HTML code, identify and explain two ways a hacker could use this page for credential harvesting.

- A fake login page can be used for credential harvesting by tricking a victim into entering their username and password. Here, following two methods are used:

Send credentials to an attacker-controlled server

- The fake page contains a login form with fields such as username/email and password.
- When the victim clicks Login, the form can submit the entered data to a remote server controlled by the attacker.
- The attacker can then store or view the submitted credentials.
- Example concept: the form’s action attribute points to an external server rather than the legitimate website.

Capture credentials locally and forward/store them

- JavaScript or server-side code can read the values entered into the form.
- The captured username and password could be saved in a file/database or transmitted elsewhere.
- The page may then redirect the victim to the genuine website making the victim think the login simply failed or that they were redirected normally.

5. Use ChatGPT or another AI tool to generate a list of 5 recent real-world hacking incidents involving Indian apps or websites. For each incident, write one line about the hacking methodology used.

No	Incident	Hacking Methodology
1	Tata electronics cyber breach - 2026	Ransomware – The world leaks group allegedly stole and published more than 630GB of Tata electronics data on the dark web.
2	Tata Technologies cyberattack - 2025	Ransomware – Attackers compromised the company network causing some IT services to be suspended.
3	Hawk Eye App - 2024	Exploitation of insecure API – Data belonging to more than 200000 users was reportedly exposed including names, phone numbers and location information.
4	boAt customer data breach - 2024	Data theft- A hacker claimed to have accessed boAt's customer database and leaked information of

		about 7.5 million customers.
5	Indian bank impersonation - 2026	Phishing – Criminals created fake banking websites/app and tricked victims into installing them and used malware to steal sensitive information from their phones.